

## LONDON LUTON, United Kingdom

WMO: 036733

Lat: 51.875N

Lon: 0.368W

Elev: 526

StdP: 14.42

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>25.2 | (c)<br>28.1 | (d)<br>22.7                   | (e)<br>17.4 | (f)<br>29.5 | (g)<br>24.7 | (h)<br>19.2 | (i)<br>30.8 | (j)<br>33.0           | (k)<br>46.6 | (l)<br>29.6 | (m)<br>46.5 | (n)<br>8.4            | (o)<br>30 | (p)<br>0.621 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.5                   | 80.8            | 66.1 | 77.0 | 64.9 | 73.5 | 63.1 | 68.3                | 77.2 | 66.4 | 74.5 | 64.4 | 71.2 | 8.6                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.8                            | 93.5 | 72.3 | 62.9 | 87.6 | 69.3 | 61.2 | 82.4 | 67.1 | 33.0          | 77.8 | 31.4 | 74.2 | 29.9 | 70.7 | 78.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 26.1          | (b) 22.8 | (c) 20.3 | (e) 22.0                   | (f) 86.9 | (g) 2.5            | (h) 3.9 | (i) 20.3                                           | (j) 89.7 | (k) 18.8   | (l) 92.0 | (m) 17.5   | (n) 94.1 | (o) 15.7   | (p) 97.0 |    |    |
| (a) 26.1          | (b) 22.8 | (c) 20.3 | (e) 21.9                   | (f) 71.8 | (g) 2.5            | (h) 2.7 | (i) 20.1                                           | (j) 73.7 | (k) 18.6   | (l) 75.3 | (m) 17.2   | (n) 76.9 | (o) 15.4   | (p) 78.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 50.3   | 39.5  | 40.1  | 43.3  | 47.7  | 53.1  | 58.7  | 63.0  | 62.6  | 57.8  | 51.7  | 44.7  | 40.5  |  |  |
|                                                                        | DBStd       | 9.99   | 5.76  | 5.78  | 5.45  | 5.53  | 5.33  | 4.91  | 4.89  | 4.71  | 4.42  | 5.11  | 5.52  | 6.39  |  |  |
|                                                                        | HDD50       | 1464   | 328   | 278   | 213   | 106   | 27    | 1     | 0     | 0     | 1     | 39    | 172   | 299   |  |  |
|                                                                        | HDD65       | 5448   | 792   | 696   | 671   | 520   | 369   | 198   | 99    | 105   | 220   | 411   | 608   | 759   |  |  |
|                                                                        | CDD50       | 1575   | 1     | 2     | 7     | 36    | 124   | 263   | 404   | 391   | 236   | 93    | 14    | 4     |  |  |
|                                                                        | CDD65       | 85     | 0     | 0     | 0     | 0     | 1     | 10    | 38    | 31    | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 694    | 0     | 0     | 0     | 2     | 9     | 102   | 316   | 232   | 30    | 3     | 0     | 0     |  |  |
|                                                                        | CDH80       | 146    | 0     | 0     | 0     | 0     | 0     | 17    | 72    | 55    | 2     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 10.3   | 11.9  | 11.8  | 11.4  | 10.0  | 9.8   | 9.3   | 9.0   | 9.0   | 9.3   | 10.0  | 10.4  | 11.4  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 27.3   | 2.6   | 1.9   | 2.1   | 2.1   | 2.0   | 2.4   | 1.8   | 2.1   | 2.4   | 2.6   | 2.6   | 2.7   |  |  |
|                                                                        | PrecMax     | 35.0   | 5.4   | 4.3   | 5.1   | 4.6   | 4.9   | 5.2   | 4.1   | 4.7   | 5.4   | 7.8   | 6.1   | 6.2   |  |  |
|                                                                        | PrecMin     | 20.3   | 0.5   | 0.3   | 0.1   | 0.2   | 0.1   | 0.2   | 0.3   | 0.1   | 0.4   | 0.2   | 0.9   | 0.6   |  |  |
|                                                                        | PrecStd     | 3.9    | 1.3   | 1.1   | 1.1   | 1.1   | 1.1   | 1.4   | 0.9   | 1.1   | 1.5   | 1.7   | 1.3   | 1.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 53.9  | 55.7  | 61.1  | 71.2  | 75.1  | 82.5  | 86.3  | 85.9  | 77.4  | 68.4  | 59.3  | 55.7  |  |  |
|                                                                        |             | MCWB   | 51.9  | 51.2  | 52.1  | 57.9  | 62.3  | 66.7  | 67.6  | 68.5  | 65.7  | 62.0  | 56.8  | 53.4  |  |  |
|                                                                        | 2%          | DB     | 52.0  | 52.1  | 57.2  | 64.5  | 71.3  | 76.8  | 81.0  | 80.4  | 72.0  | 64.1  | 57.2  | 53.8  |  |  |
|                                                                        |             | MCWB   | 50.0  | 49.1  | 50.8  | 55.0  | 61.0  | 64.6  | 65.7  | 67.0  | 63.0  | 59.2  | 54.7  | 51.9  |  |  |
|                                                                        | 5%          | DB     | 50.0  | 50.2  | 53.9  | 60.6  | 67.6  | 72.0  | 77.1  | 75.4  | 68.3  | 61.0  | 55.2  | 52.0  |  |  |
|                                                                        |             | MCWB   | 48.5  | 48.0  | 49.2  | 52.9  | 58.7  | 62.1  | 64.6  | 64.7  | 60.8  | 57.7  | 53.0  | 50.0  |  |  |
|                                                                        | 10%         | DB     | 48.2  | 48.3  | 51.8  | 57.1  | 62.9  | 68.4  | 73.4  | 71.7  | 65.9  | 59.2  | 53.3  | 50.0  |  |  |
|                                                                        |             | MCWB   | 47.0  | 46.7  | 47.9  | 50.8  | 55.7  | 60.0  | 63.1  | 62.9  | 59.1  | 56.3  | 51.2  | 48.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 52.9  | 52.3  | 54.0  | 59.1  | 65.4  | 69.1  | 70.0  | 72.5  | 67.9  | 62.7  | 57.7  | 54.4  |  |  |
|                                                                        |             | MCDB   | 54.0  | 55.2  | 58.5  | 68.4  | 71.6  | 78.9  | 80.5  | 82.6  | 74.3  | 67.5  | 59.2  | 55.7  |  |  |
|                                                                        | 2%          | WB     | 50.7  | 50.3  | 52.1  | 55.9  | 61.8  | 66.0  | 67.9  | 68.4  | 64.7  | 60.3  | 55.4  | 52.8  |  |  |
|                                                                        |             | MCDB   | 51.9  | 51.9  | 56.1  | 63.2  | 68.7  | 74.3  | 77.8  | 76.7  | 70.5  | 63.2  | 56.9  | 54.2  |  |  |
|                                                                        | 5%          | WB     | 48.6  | 48.4  | 50.5  | 53.5  | 59.3  | 63.4  | 65.9  | 66.0  | 62.5  | 58.8  | 53.3  | 50.9  |  |  |
|                                                                        |             | MCDB   | 49.5  | 49.7  | 53.4  | 59.1  | 65.5  | 70.0  | 74.8  | 73.1  | 67.1  | 61.0  | 55.0  | 52.2  |  |  |
|                                                                        | 10%         | WB     | 46.7  | 46.7  | 48.8  | 51.3  | 57.0  | 61.2  | 63.9  | 63.9  | 60.3  | 57.1  | 51.5  | 48.7  |  |  |
|                                                                        |             | MCDB   | 47.8  | 48.2  | 51.3  | 56.1  | 62.2  | 67.4  | 71.3  | 69.8  | 64.2  | 59.1  | 53.2  | 49.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.5   | 9.6   | 12.2  | 14.8  | 15.5  | 16.1  | 16.5  | 15.7  | 14.0  | 11.5  | 9.1   | 8.0   |  |  |
|                                                                        |             | MCDBR  | 10.2  | 10.8  | 15.5  | 20.2  | 21.6  | 22.1  | 23.2  | 21.6  | 17.9  | 12.8  | 9.6   | 9.1   |  |  |
|                                                                        |             | MCWBR  | 10.0  | 8.8   | 10.6  | 12.3  | 12.9  | 12.2  | 10.7  | 10.6  | 10.1  | 9.1   | 8.8   | 9.0   |  |  |
|                                                                        | 5% WB       | MCDBR  | 10.2  | 10.3  | 14.3  | 18.0  | 19.1  | 20.0  | 20.7  | 18.8  | 16.2  | 11.7  | 9.4   | 9.1   |  |  |
|                                                                        |             | MCWBR  | 10.3  | 9.2   | 10.7  | 11.7  | 12.4  | 12.5  | 11.3  | 11.0  | 10.4  | 9.3   | 9.1   | 9.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.321 | 0.337 | 0.371 | 0.393 | 0.398 | 0.399 | 0.396 | 0.386 | 0.368 | 0.351 | 0.325 | 0.311 |  |  |
|                                                                        | taud        |        | 2.378 | 2.391 | 2.289 | 2.242 | 2.264 | 2.285 | 2.301 | 2.351 | 2.404 | 2.438 | 2.424 | 2.359 |  |  |
|                                                                        | Ebn at Noon |        | 216   | 247   | 259   | 267   | 271   | 271   | 270   | 266   | 258   | 239   | 213   | 198   |  |  |
|                                                                        | Edh at Noon |        | 19    | 25    | 33    | 39    | 40    | 40    | 39    | 35    | 30    | 24    | 18    | 17    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 251   | 463   | 805   | 1244  | 1514  | 1655  | 1576  | 1308  | 984   | 572   | 302   | 199   |  |  |
|                                                                        | RadStd      |        | 22    | 60    | 104   | 140   | 99    | 162   | 161   | 101   | 85    | 51    | 34    | 23    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (84 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page